

EXAM 7 – SPRING 2011

14. (2 points)

A firm is incorporating extreme events into an enterprise risk model. Discuss the difficulty in producing accurate results for the model when considering:

a. (0.5 point)

Catastrophe risks.

b. (0.5 point)

Operational risk loss data.

c. (0.5 point)

Expert opinion.

d. (0.5 point)

The causes of extreme events.

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Sample Solutions
Exam 07, 2011, Question 14

Solution 1

a)

Catastrophes are very infrequent so few historical data points are available. Every future catastrophe will affect the model and its subsequent predictions, and there is considerable variability in possible loss amounts.

Any 2 of the following examples:

- The degree of detail and research that is put into separate cat models varies significantly so some models are more accurate than others.
 - The assumed frequency of occurrence of the modeled cat event is usually entirely based on judgment.
 - These models often draw on extremely sparse data to determine the potential magnitude of a catastrophic event.
 - After determining potential magnitude of event, cat models usually attempt to estimate the impact of such an event allowing for changes in conditions (such as national response plans, medical advances, changes in social habits, changes in building techniques etc). This involves judgment.
- b) Operational risk loss data is often taken from third party databases and may, for example, include extreme loss events experienced by other industries (the banking industry for example). These would then be scaled to a size appropriate to the insurer conducting the modeling by using reference metrics; this approach introduces its own risk. Other appropriate answers may be accepted.
Operational risk extreme events are also often developed solely based on discussion among experts within a business, who naturally draw on their own experiences and knowledge of the experiences of other similar companies. This is subject to the biases inherent in the use of judgment.
- c) Using experts within a business involves drawing on their own experiences and knowledge of other similar companies. This is subject to the problems inherent in the use of judgment -for example their knowledge may be about different types of firms, they may have limited information, their judgment may be biased, etc..
- d) The analysis undertaken, and the model design for extreme events, often focuses on the outcomes of a risk rather than the causes. The models generally make limited or no adjustment for the possible evolution of the risk environment such as
- increased globalization
 - growth of leverage
 - increased/decreased regulation/supervisory oversight